

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

The area of a rectangular region is increasing at a rate of **250** square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use **1 meter = 3.28 feet**.)

Answer

- A. **0.39**
- B. **1.27**
- C. **13.67**
- D. **23.24**

Correct Answer: A

Rationale

Choice A is correct. It’s given that **1 meter = 3.28 feet**. It follows that **1² square meter = 3.28² square feet**, or **1 square meter = 10.7584 square feet**. Since **1 hour = 60 minutes**, it follows that **250** square feet per hour is equivalent to $\left(\frac{250 \text{ square feet}}{1 \text{ hour}}\right)\left(\frac{1 \text{ square meter}}{10.7584 \text{ square feet}}\right)\left(\frac{1 \text{ hour}}{60 \text{ minutes}}\right)$, or $\frac{250 \text{ square meters}}{645.504 \text{ minutes}}$, which is approximately **0.3873** square meters per minute. Of the given choices, **0.39** is closest to **0.3873**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.